

GM-DTS-04/08 Distributed Temperature Interrogator

Distributed temperature sensing host · Raman scattering + OTDR · fire-system interlock

In-house temperature-stabilized optoelectronic module and fully embedded board (ARM + Linux); classified as a linear heat fire detector under the code, one unit covers several kilometres and locates to within a metre, with long service life and no maintenance. Alarms connect directly to fire panels, fans and pumps via RS485 / dry contacts.



GM-DTS-04/08 interrogator

Cable tunnels

Utility corridors

Road tunnels

Dams

Oil & gas storage fire warning

Features

- 0.2 m spatial resolution, locating to within a metre
- $\pm 1\text{ }^{\circ}\text{C}$ accuracy, $0.1\text{ }^{\circ}\text{C}$ resolution
- 0.5 s measurement time (10,000 accumulations)
- 12.5 km per channel; 4 channels standard, 8 optional
- Temperature-stabilized optics for minimal long-term drift
- ARM + Linux embedded board, no fan, no hard disk
- RS485 / dry contacts direct to fire panel, fan and pump interlock
- Complies with GB 50116 · GB 50838 · GB/T 21197

Specifications

Sensing performance

Fiber type	Multimode (single-mode long range on request)
Measurement range	12.5 km / channel
Channels	4 (8 optional)
Spatial resolution	0.2 m
Accuracy / resolution	$\pm 1\text{ }^{\circ}\text{C}$ / $0.1\text{ }^{\circ}\text{C}$
Measurement time	0.5 s (10,000 accumulations)
Temperature range	-20~120 $^{\circ}\text{C}$

Alarming & interfaces

Electrical interfaces	RS485 / RJ45 / CAN
Fire interlock	Dry contacts direct to fire panel / fans / pumps

Power & mechanical

Power / consumption	85~264 V AC · 15~25 W
Mounting	2U rack mount

Ordering information

Model	Configuration	Notes
GM-DTS-04	4 channels	Standard unit, 4×12.5 km per box
GM-DTS-08	8 channels	Wide-coverage unit
Custom	Single-mode long range	Ultra-long-distance sensing per project

Typical configuration

1 km fire compartment = one GM-DTS-04/08 + sensing cable (sinusoidal surface run / suspended) + zoning software; add GM-FT02 fast-response units at local high-risk points.

Warranty & service

2 yr Hardware warranty · spares stocked	24 h On-site response, East China	48 h On-site response, nationwide	7×24 Remote diagnostics hotline
---	---	---	---

Standard service policy; the supply contract prevails.

Certifications & standards

 China National Accreditation Service for Conformity Assessment CNAS testing	 计量认证 CMA metrology	 ISO 9001 quality	 ISO 14001 environmental	 矿用产品安全标志 MA mining safety
---	--	---	--	---

Certificate numbers and scans are available on request.